HR Analytics Dashboard

PRDA-03 · Workforce overview, attrition analysis and recommendations

Power BI · Tableau · MySQL · Excel

About the dataset

- Source: MySQL database project_hr_analytics, table HR
- One row per employee: name, birthdate, age, gender, race, department, job title
- Location (Headquarters or Remote), city and state
- hire_date and termdate (blank if still employed)
- Goal: understand the workforce and why people leave

22,214
employees

13
columns

2000–2020
hire dates

Data cleaning and preparation

Loaded all rows

Removed the 1,000-row limit: 22,214 rows, no duplicate IDs

Fixed date formats

hire_date and birthdate read as day-month-year; termdate trimmed to the date only

Handled missing termdate

18,285 blanks and 1,030 future dates treated as Active employees

Added new columns

Employee Status, Age Group, Hire Year, Term Year, Tenure, Tenure Group

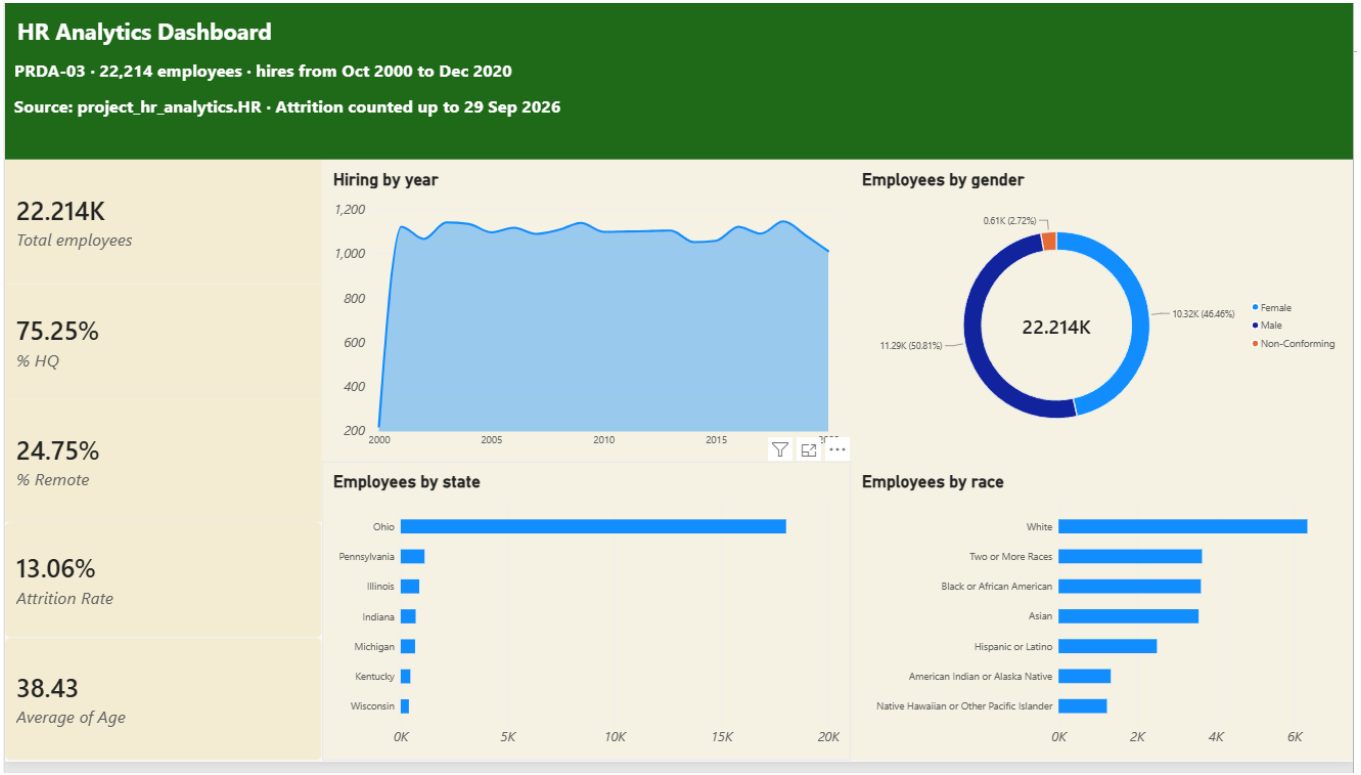
Key performance indicators

22,214	75.2%	24.8%	38.4	13.1%
Total employees	At headquarters	Working remotely	Average age (years)	Attrition rate

19,313 active employees · **2,901** have left · leavers stayed **8 years** on average

Attrition counts termdates up to the date of analysis, so it can differ by 0.01% between tools.

Power BI dashboard: workforce overview



Power BI dashboard: departments and attrition

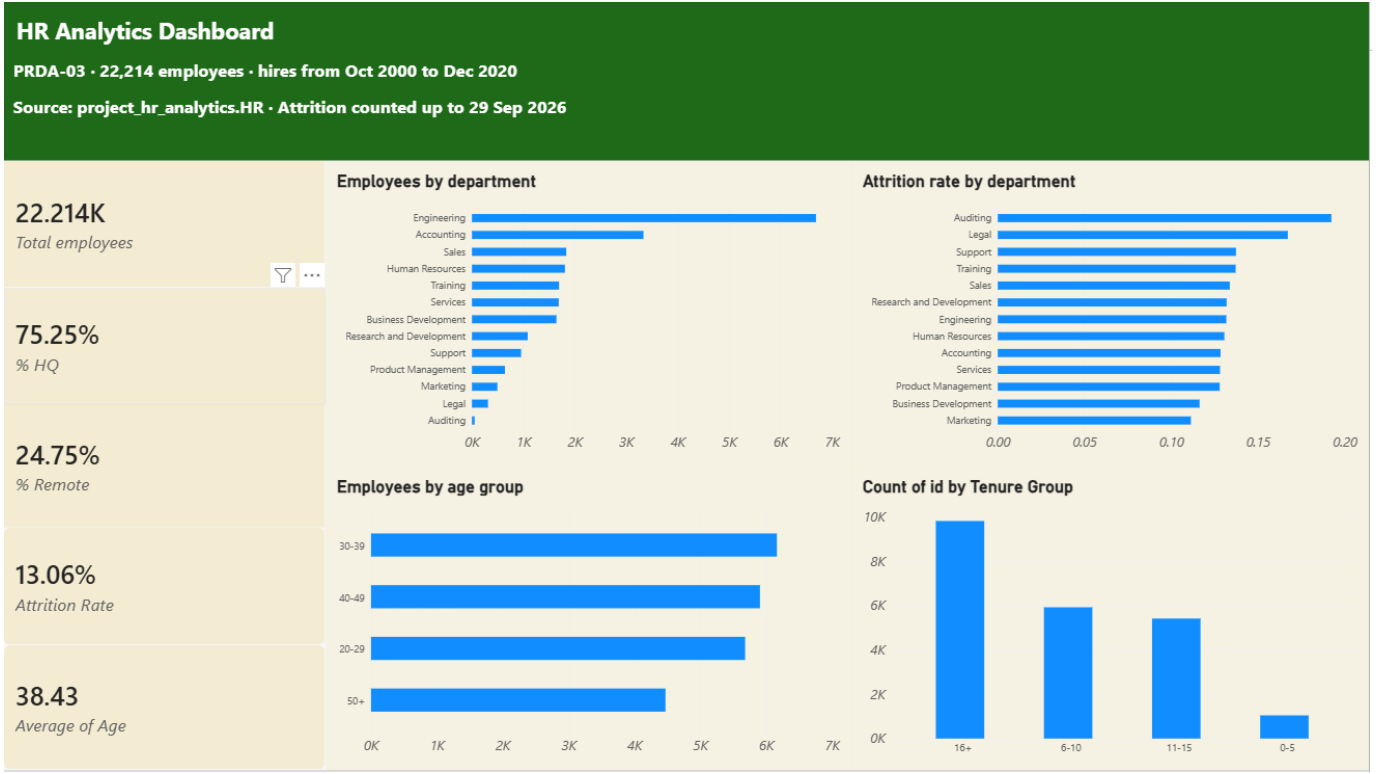
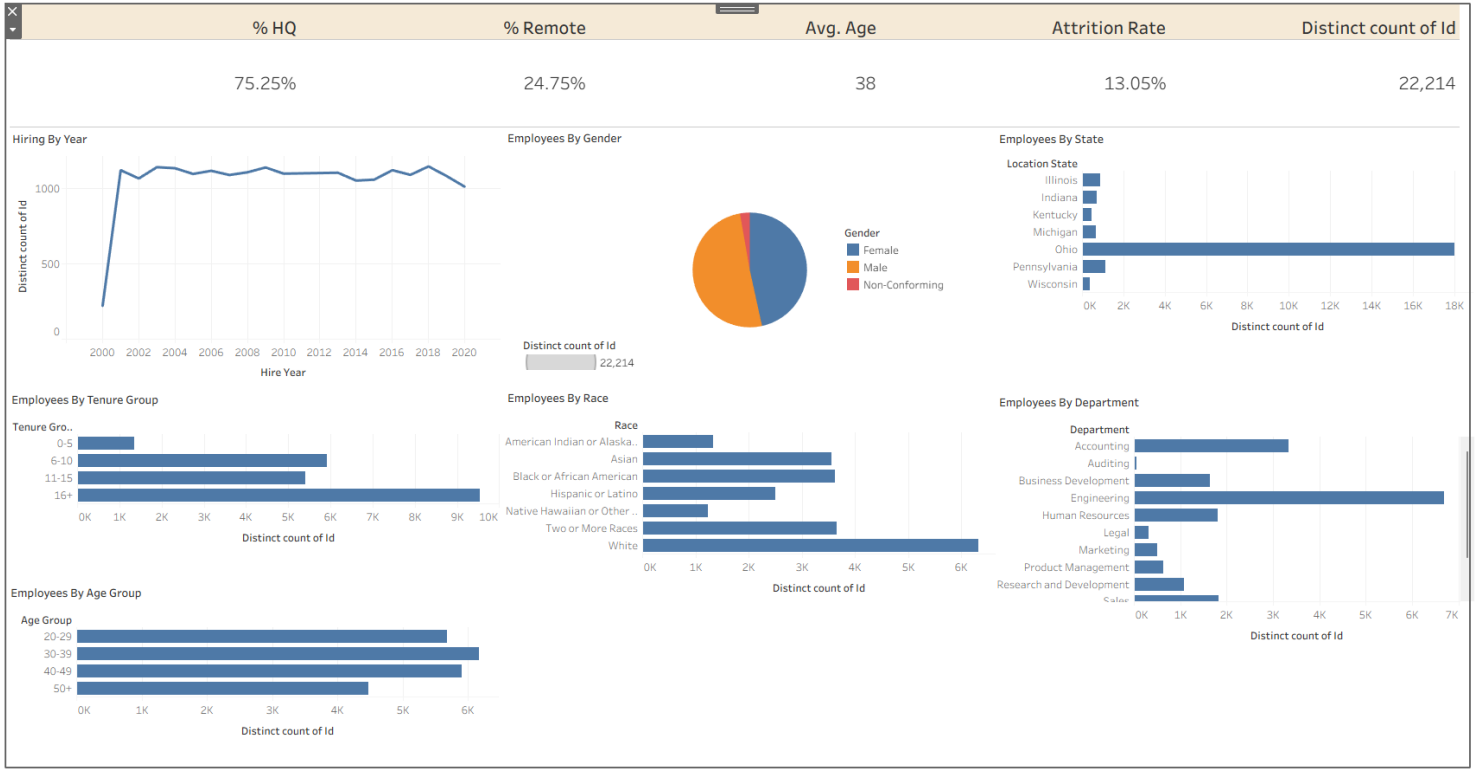
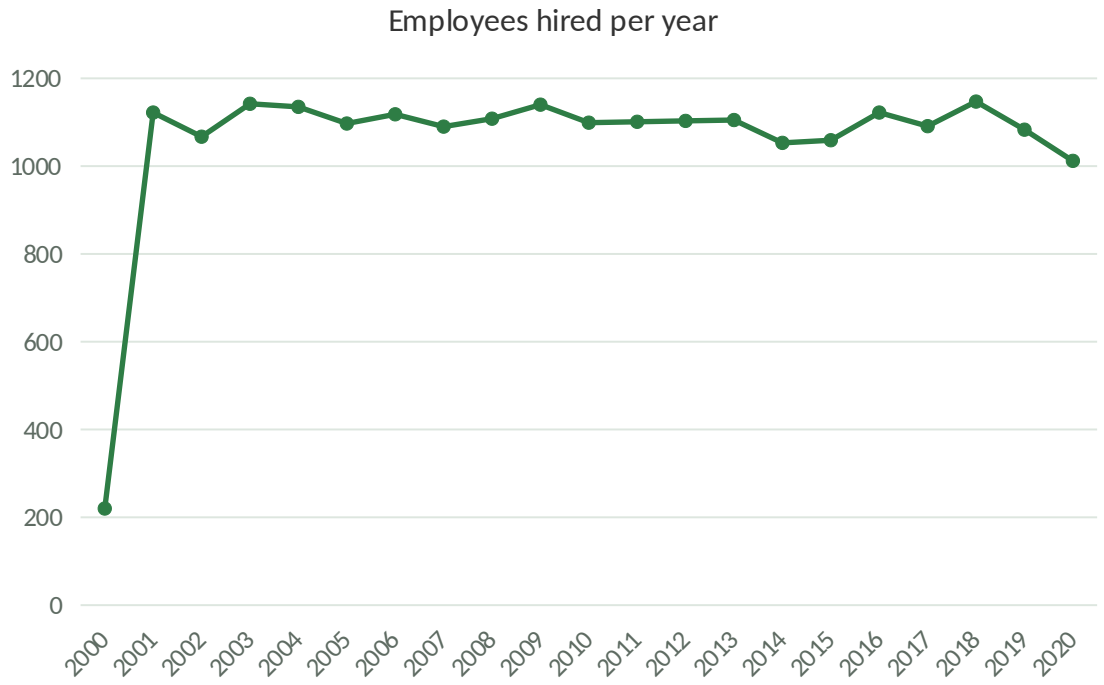


Tableau dashboard



Hiring has been steady for 20 years

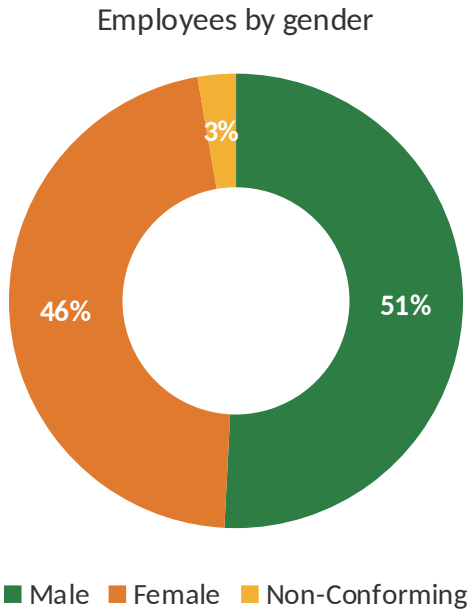


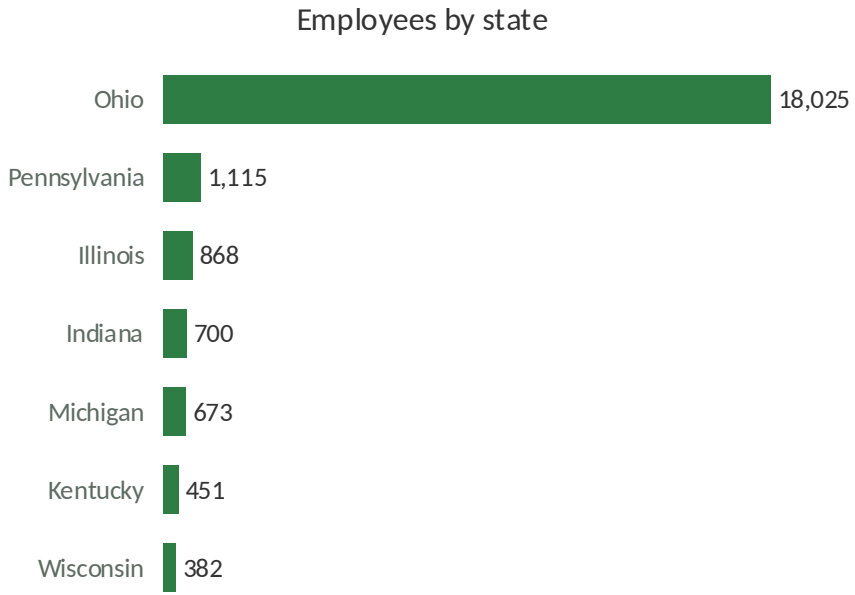
≈1,100

hires every year, 2001–2020

No big growth or freeze. 2000 is low (220) only because the data starts in October 2000.

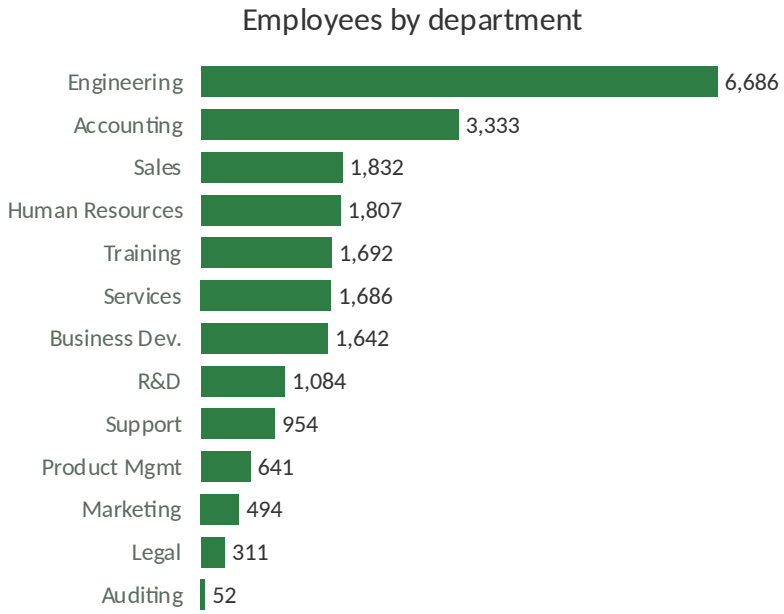
Balanced gender, concentrated in Ohio

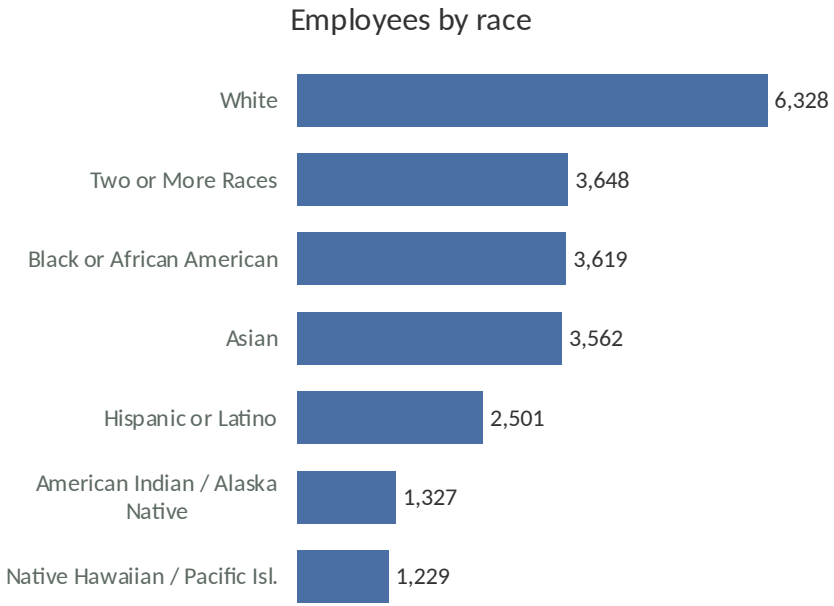




Women are 43–48% of every department. 81% of employees work in Ohio.

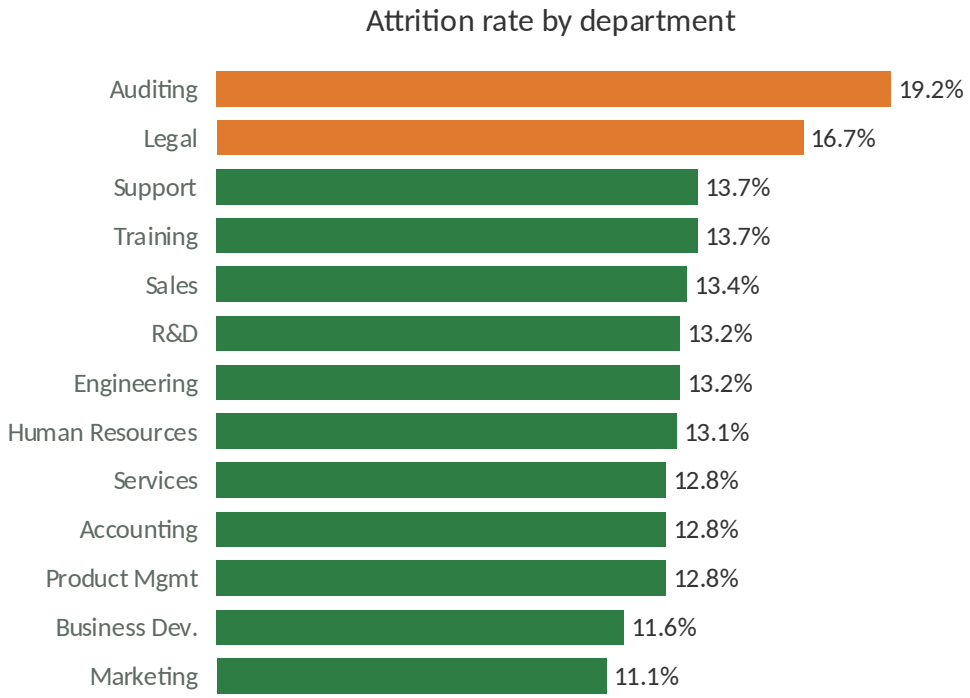
Engineering and Accounting are half the company





Engineering (30%) and Accounting (15%) are the core teams. No race group other than White is above 17%.

Legal and Auditing lose the most staff



19.2%

Auditing attrition

Legal follows at 16.7%. Auditing has only 52 people, so Legal (311 people) is the more reliable warning sign. Most teams sit near 13%.

Leavers keep rising each year



7 → 205

leavers in 2001 vs 2021

Hiring stayed flat at about 1,100 a year while leavers grew every year, so net growth is slowing. 2026 covers January to September only.

Most leavers go in their first 5 years

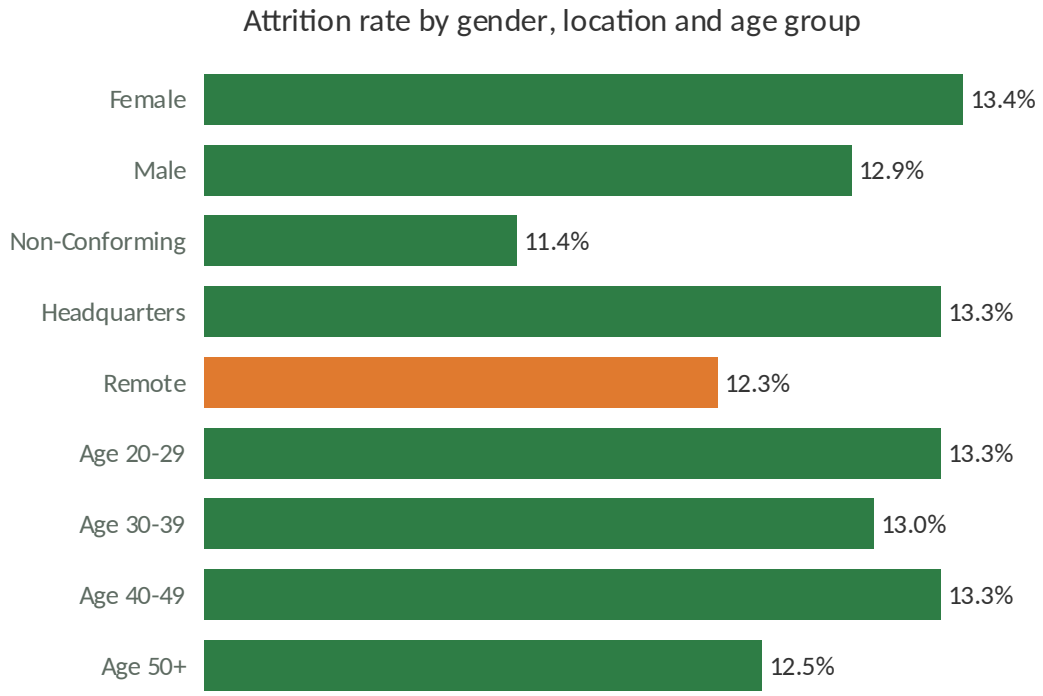


39%

of leavers left within 5 years

The longer people stay, the less likely they are to leave. Early-career retention is the biggest opportunity.

Attrition is similar across groups



12.3%

remote attrition (HQ 13.3%)

Remote employees stay slightly longer. Gender, age and race all sit between 11% and 15%, so no group leaves at a much higher rate.

Key insights

Steady hiring

About 1,100 hires a year for 20 years

Balanced and diverse

50.8% male, 46.5% female, 7 race groups

One main location

81% in Ohio, 75% at headquarters

Rising attrition

13.1% have left; leavers grow every year

Early leavers

39% of leavers go within 5 years

Hot spots

Legal 16.7% and Auditing 19.2% attrition

Suggestions for the company

1. **Improve early-career retention**
   Better onboarding, mentors and regular check-ins

2. **Investigate Legal and Auditing**
   Exit interviews and engagement surveys

3. **Plan hiring against attrition**
   Track hires minus leavers every quarter

4. **Support flexible work**
   Remote staff leave less: offer hybrid options

5. **Prepare for retirements**
   20% are 50+: start succession planning

6. **Widen the location base**
   Grow branches and remote hiring beyond Ohio

7. **Keep diversity strong**
   Monitor gender and race balance every year

8. **Improve HR data quality**
   Fix future termdates and add hires after 2020

Conclusion

A large, diverse and stable workforce, but attrition is rising and most people leave early. Focusing on early-career retention and the high-attrition teams will keep the company growing.